

VOLUME 10

COMPLETE VOLUME — CHAPTERS 10.1–10.7

Deployment

Releasing the Application — the Final Volume in the Documentation Set

Document Title	Volume 10 — Deployment (Combined)
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Governed By	Volume 0 — Project Foundation; grounded in Volumes 6, 7, 8, and 9

Table of Contents

This volume releases the application: Android and iOS build/signing, the store submission checklists (kept consistent with Volume 8's privacy disclosures), production monitoring against Volume 0's own success metrics, crash-reporting operations built on Volume 6's ADR-010, and the single end-to-end release runbook tying every gate together.

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CHAPTER 10.1

Android Release

Signing, Versioning, and Building the Primary Platform's Release Artifact

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1. Build Artifact

An Android App Bundle (.aab), not a universal .apk — Play Store's required upload format since it lets Google Play generate optimized, device-specific APKs rather than shipping every device a one-size-fits-all binary.

2. Signing

Play App Signing (Google-managed upload key): the project's own upload key signs the .aab for upload; Google re-signs it with the actual distribution key it manages — meaning a lost local upload key is recoverable (Google can help re-establish a new one), unlike a fully self-managed signing key with no recovery path.

Upload keystore storage: generated once, stored in a password-protected keystore file, itself stored in the CI secret store (Volume 7, Chapter 7.8/7.13) — never committed to the repository, per Volume 3, Chapter 3.7's secret-handling rule.

3. Versioning

Field	Scheme	Example
versionName	Semantic (MAJOR.MINOR.PATCH), human-facing	1.2.0
versionCode	Monotonically incrementing integer, required by Play Store, never reused	14

Both are bumped together as part of Chapter 10.7's release process, driven from a single source of truth (pubspec.yaml's version field, per Flutter convention) rather than edited independently in two places.

4. Build Flavors at Release Time

Only the prod flavor (Volume 7, Chapter 7.11) is ever built for Play Store upload — pointing at the production Firebase project (Chapter 7.7) and production API base URL (Chapter 7.10); dev/staging flavors are never accidentally shippable since they use entirely separate Firebase projects and app IDs, making a wrong-flavor upload immediately obvious in the Play Console.

5. Required Permissions Declared

- Camera, Microphone, Location (foreground, during active recording only), Storage, and Notifications — each one justified by a specific onboarding explanation already designed in Volume 2, Chapter 2.7 (C-01) and required by Google Play's permissions-justification policy, detailed further in Chapter 10.3.

6. What This Chapter Defers

- Store listing content and data-safety disclosures — Chapter 10.3.
- The full release runbook this build step fits into — Chapter 10.7.

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CHAPTER 10.2

iOS Release

The Secondary-Priority Platform, Released via the Cloud Mac Pipeline

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Reflects	Volume 0, Chapter 0.2's iOS-secondary-priority stance

1. Build & Signing — Entirely on the Cloud Mac Service

Consistent with every prior volume's treatment of iOS: build, code-signing, and archive creation happen on the cloud Mac / CI service configured in Volume 7, Chapters 7.1 and 7.8 — this Windows-based workflow never touches Xcode or a local certificate directly.

- Distribution certificate and provisioning profile are managed through Apple's automatic signing (via App Store Connect API keys, stored as CI secrets per Volume 7, Chapter 7.13) rather than manually managed .p12/.mobileprovision files passed around outside CI.

2. Versioning

CFBundleShortVersionString and CFBundleVersion mirror Android's versionName/versionCode exactly (Chapter 10.1) — both platforms release under the same version number for the same underlying build, so "v1.2.0" always means the identical feature set on both platforms, avoiding a confusing cross-platform version mismatch.

3. TestFlight Before App Store

Every release candidate goes to TestFlight first: internal testers (the project owner and any designated reviewers) validate the build via TestFlight before it's submitted for App Store review — mirroring Volume 9, Chapter 9.8's UAT sign-off gate, run once per platform.

4. Reduced Cadence, Not Reduced Rigor

Volume 0's iOS-secondary stance means iOS releases may trail Android releases in timing (e.g. shipped a few days later, once the cloud Mac pipeline completes and TestFlight review clears) — it does not mean iOS skips any gate Chapter 9's testing pyramid or Chapter 10.7's release process requires. The same device-matrix and manual-testing bar (Volume 9, Chapters 9.8/9.9) applies, just against a smaller iOS device set.

5. What This Chapter Defers

- App Store Connect listing content and the App Privacy nutrition label — Chapter 10.4.
- The shared release runbook both platforms follow — Chapter 10.7.

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CHAPTER 10.3

Play Store Checklist

Everything Google Play Requires Before This App Can Go Live

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Chapter Reference	10.3
Version	1.0
Status	Draft — Pending Approval
Owner	Faisal — Human Archive
Prepared With	Claude (Cowork)
Date	05 August 2026
Must Match	Volume 8, Chapter 8.6 (Privacy Policy) exactly

1. Store Listing

- App name, short/full description, screenshots — screenshots are taken from the actual built app (not the Chapter 2.6 wireframes) once available, using the Chapter 2.8 Design System's real visual treatment.
- Feature graphic and icon follow the brand colors already fixed in Volume 2, Chapter 2.8, Section 1 — visual consistency between the shipped app and its store presence.

2. Data Safety Section

Must declare exactly the data collection already specified in Volume 8, Chapter 8.6 — no more, no less. A mismatch between what's declared here and what the app actually collects is both a policy violation and a trust problem worth avoiding by construction, not just by careful form-filling.

Data Type	Declared As	Purpose
Location (precise)	Collected, not shared with third parties beyond Chapter 8.6's named processors (AWS)	App functionality (FR-META-05).
Photos/videos	Collected	App functionality (core product).
Personal info (name, email)	Collected	Account management.
App activity / device identifiers	Collected	App functionality, analytics (Volume 9, Chapter 9.3).

3. Permissions Declarations

Each permission from Volume 10, Chapter 10.1, Section 5 requires an in-Play-Console justification; the justification text is drawn directly from Volume 2, Chapter 2.7's C-01 onboarding copy, since Google's review reads more favorably on permission requests that are already clearly explained to the end user in-app.

4. Content Rating

Completed via Play Console's rating questionnaire — expected to land in the lowest/most general category, since this is a business-to-business field-data-collection tool with no user-generated public content, violence, or age-sensitive material.

5. Privacy Policy Link

A published, hosted version of Volume 8, Chapter 8.6 (after the legal review that chapter itself requires) is linked in the Play Console listing — this is a hard Play Store submission requirement, not optional, and the link must resolve before submission, not be added after the fact.

6. Staged Rollout

- Initial release goes out to a small percentage (e.g. 10–20%) of eligible users/devices via Play Console's staged rollout, expanded to 100% only after Chapter 10.5's monitoring shows no elevated crash rate or upload failure rate versus the previous version — the concrete mechanism behind Chapter 10.7's rollback-friendly release philosophy.

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CHAPTER 10.4

App Store Checklist

Apple's Equivalent Requirements — and Where They're Stricter

Document Title	App Store Checklist
Chapter Reference	10.4
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Must Match	Volume 8, Chapter 8.6 (Privacy Policy) exactly

1. App Privacy 'Nutrition Label'

App Store Connect's data-collection disclosure is stricter and more granular than Play's Data Safety section (Chapter 10.3) — each data type must be tied to a specific purpose (App Functionality, Analytics, etc.) individually, matching Volume 8, Chapter 8.6's data table exactly, entry for entry.

2. Permission Usage Strings (Info.plist)

Key	Required Justification Text (Drawn From Ch.2.7's C-01 Copy)
NSCameraUsageDescription	"Camera access is needed to record your assigned walkthroughs."
NSMicrophoneUsageDescription	"Microphone access is needed to record audio alongside your video."
NSLocationWhenInUseUsageDescription	"Location is recorded at the moment each video segment is captured, to record where the task was performed."
NSUserNotificationsUsageDescription (implicit via permission prompt)	Justified via in-app priming (C-01) before the OS prompt, per Volume 2, Chapter 2.9.

Location string specificity matters most here: Apple's App Review scrutinizes location-usage strings closely — a vague string is a common rejection reason. This one deliberately states the exact, narrow, task-tied purpose (matching Volume 8, Chapter 8.6, Section 2's stated rationale precisely) rather than a generic "to improve your experience."

3. Background Modes Declaration

The background URLSession capability (Volume 5, Chapter 5.11; Volume 3, ADR-003) must be declared in the app's background modes — Apple's review specifically checks that a declared background capability is genuinely used for its stated purpose (resumable upload) and not for something undisclosed, which this app satisfies directly since that is its only background use.

4. Review Notes for Apple's Reviewer

- A brief reviewer note explains the app's actual context (a B2B field-data-collection tool, Admin/Collector accounts provisioned by the client organization, not public self-signup) — reducing the chance of a reviewer misunderstanding the login flow (SH-02) as a consumer app missing a "Sign Up" option, since there deliberately isn't one (FR-ADM-07).
- Reviewer test credentials (a demo Admin and Collector account) are provided, since App Review has no other way to get past Login otherwise.

5. What This Chapter Defers

- TestFlight process before submission — Chapter 10.2.
- The shared release runbook — Chapter 10.7.

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CHAPTER 10.5

Monitoring

Watching the Vision Document's Own Metrics in Production

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Resolves	The operational half of Volume 6, Chapter 6.9's deferred alerting-threshold item

1. Dashboards

Dashboard	Source	Audience
Vision Document KPIs	Volume 9, Chapter 9.3's queries (upload success rate, data-loss incidents, latency, recovery time)	Project owner — direct read-out against Volume 0's original success metrics.
Infrastructure health	CloudWatch (Lambda error rate/duration, Aurora CPU/connections, API Gateway 4xx/5xx rate)	Whoever is on operational duty.
Crash-free rate	Firebase Crashlytics (Volume 6, Chapter 6.9, ADR-010)	Same audience; detailed further in Chapter 10.6.

2. Alerting Thresholds

Signal	Threshold	Action
Upload success rate (weekly)	Drops below 95%	Investigate before it becomes a Collector-visible trend; check Volume 5, Chapter 5.13's retry telemetry first.
API 5xx rate	> 1% of requests over a 15-minute window	Page whoever's on operational duty — likely a backend (Volume 4) regression.
Crash-free users	Drops below 99% for the current release	Evaluate against Chapter 10.6's release-blocking threshold; may trigger a rollback (Section 3).
Aurora CPU/connection saturation	Sustained > 80%	Check Volume 4, Chapter 4.1's Aurora Serverless v2 auto-scaling is actually keeping up; escalate capacity if not.

3. Staged Rollout Monitoring Window

During Chapter 10.3's staged rollout, the dashboards in Section 1 are watched specifically for any regression versus the previous release's baseline — a hard rule, not a suggestion: rollout does not advance past its initial percentage while any Section 2 threshold is breached for the new version specifically.

4. What This Chapter Defers

- Crash-specific triage process — Chapter 10.6.
- The full release runbook these gates plug into — Chapter 10.7.

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CHAPTER 10.6

Crash Reporting

Operationalizing ADR-010's Crashlytics Choice

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Builds On	Volume 6, Chapter 6.9, ADR-010 (Firebase Crashlytics)

1. Crash-Free-Users Target

99.5% crash-free users per release — a deliberately high bar, since a crash mid-recording risks the one thing the entire product exists to prevent: lost field footage. This target directly serves Volume 0's own Vision Document goal of near-total upload reliability and zero data-loss incidents.

2. Triage Process

- Every new crash cluster (grouped by Crashlytics automatically) is triaged within one business day of first appearing in a release build — not necessarily fixed immediately, but acknowledged and severity-assessed.
- A crash occurring during the Recording or Upload flows (Volume 5) is treated as higher severity than one in, say, the Settings screen, given the core-promise reasoning in Section 1 — triage severity is risk-weighted by which flow is affected, not just by raw frequency.
- Non-fatal errors logged per Volume 6, Chapter 6.9 (e.g. a chunk hitting its final retry attempt) are reviewed weekly for emerging patterns, distinct from the daily fatal-crash triage.

3. Release-Blocking Threshold

A release is rolled back (Chapter 10.3's staged rollout, halted and reverted) if: crash-free rate for the new version drops more than 0.5 percentage points below the previous version's crash-free rate at an equivalent rollout percentage — a relative regression check, not just an absolute number, so a naturally noisy small-sample early rollout doesn't trigger a false rollback.

4. Symbolication Across Platforms

- Android: Crashlytics automatically de-obfuscates using the uploaded mapping file from the release build (Chapter 10.1).
- iOS: dSYM files from the cloud Mac build (Chapter 10.2) are uploaded to Crashlytics as part of the same CI job that produces the archive, so no iOS crash ever sits un-symbolicated waiting on a manual upload step.

5. What This Chapter Defers

- The full release process this triage/rollback logic plugs into — Chapter 10.7.

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CHAPTER 10.7

Release Process

The End-to-End Runbook Every Chapter in This Volume Feeds Into

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Chapter Reference	10.7
Version	1.0
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1. Purpose

Every prior chapter in Volumes 9 and 10 built one piece of the release gate — this chapter is the single ordered runbook tying them together, so "ship a release" is a checklist to follow, not a set of scattered facts to remember.

2. The Runbook

1. Version bump: increment pubspec.yaml's version (Chapter 10.1/10.2's shared versioning scheme); write the changelog entry.
2. Full test suite passes: Volume 9, Chapters 9.6/9.7's unit and integration suites green in CI (Volume 7, Chapter 7.13).
3. Device matrix passes: Volume 9, Chapter 9.9's Android/iOS device matrix, run manually against the release candidate build.
4. UAT sign-off: the project owner (or designated Admin-role tester) completes Volume 9, Chapter 9.8's manual test plan and gives explicit approval — mirroring Volume 0's own approval discipline, now applied to a release rather than a document.
5. Build release artifacts: Android .aab (Chapter 10.1) and iOS archive via the cloud Mac pipeline (Chapter 10.2), both from the exact same tagged commit.
6. Submit: Play Console (Chapter 10.3) and App Store Connect (Chapter 10.4), including the reviewer notes/test credentials Chapter 10.4 specifies for Apple.
7. Staged rollout: begin at the initial percentage (Chapter 10.3); watch Chapter 10.5's dashboards continuously against Section 3's thresholds.
8. Expand or roll back: advance to 100% only once the monitoring window (Chapter 10.5) shows no regression; roll back immediately per Chapter 10.6's release-blocking threshold if it does.
9. Post-release monitoring window: dashboards (Chapter 10.5) are checked at least daily for the first week after reaching 100%, then fold back into standard ongoing monitoring cadence.

3. Rollback Plan

Android: Play Console supports halting a staged rollout instantly and, if needed, releasing a previous version as a new, higher-versionCode build (Android does not support true downgrade) — this is why the rollback decision (Chapter 10.6) is made early in a rollout, while the blast radius is still small.

iOS: Apple does not support halting App Review-approved rollout the same way; the practical rollback is expediting a follow-up fix release through App Review, which is part of why Chapter 10.2's TestFlight gate and this chapter's device-matrix/UAT gates exist — to catch problems before submission, since post-submission rollback is slower on this platform.

4. Who Can Approve Each Gate

- Steps 4 (UAT sign-off) and 9 (rollback decision) require the project owner's explicit approval — every other step is a mechanical, automatable gate (Volume 7, Chapter 7.13's CI) that either passes or blocks without needing a judgment call.

5. Volume 10 — Closing Note

This is the last of the 13 planned volumes (Volume 0 through Volume 10, plus the phased implementation work Volume 1's roadmap already described). Per Volume 0's own Constitution — no code before every foundation, product, design, architecture, backend, recording-engine, mobile-architecture, environment, security, and QA volume is reviewed and approved — this release process is the final piece the whole documentation set has been building toward.